

# Amos Maponya

## Data Scientist

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Location: Bergbron, Randburg

[LinkedIn](#) [GitHub](#)

### Profile

Data Scientist with an Honours degree in Statistics and hands-on experience across the full data life-cycle from data pipeline development to machine learning model deployment and business intelligence reporting. Experienced working with sensitive financial and personal data in regulated environments, building automated pipelines, compliance-driven reporting systems, and API integrations that replaced manual processes with reliable, monitored workflows. Skilled in SQL, Python, and Power BI, with experience across insurance, credit, and enterprise data environments.

### Education

**ExploreAI Academy**, Higher System Development (Data Science) Jan 2024– Aug 2024

- GPA: 77%
- **Coursework:** Preparing Data (Data Cleaning and Data Preprocessing), SQL, Data Visualisation, Python, Regression, Natural Language Processing (NLP), Classification, Unsupervised Learning.

**University of Limpopo**, BSc Honours in Statistics Jan 2021 – Nov 2021

- GPA: 61%
- **Coursework:** Probability Theory, Statistical Inference, Multivariate Analysis, Categorical Data Analysis, Research Project (Statistics)

**University of Limpopo**, BSc in Mathematics and Statistics Jan 2016 – Jun 2020

- GPA: 60%
- **Coursework:** Mathematical Analysis I, Abstract Algebra, Mathematical Analysis II, Complex Analysis, Time Series Analysis, Applied Linear Regression, Sampling Theory, Design and Analysis of Experiments

### Experience

**Data Scientist | Full-time**, ADEI (PTY) Ltd Dec 2025 – Present

- Built and deployed WhatsApp chatbot solutions for debt collection companies, micro-lenders, and municipalities, automating customer engagement and payment workflows.
- Integrated bots with banking APIs, third-party bill payment gateways, and caller/voice systems, enabling end-to-end automated payment and communication flows.
- Designed and maintained data pipelines to capture, validate, and store chatbot and integration data for analytics and compliance reporting.
- Monitored production bots and integrations to proactively detect failures and anomalies, minimizing downtime and client impact.
- Developed performance dashboards tracking bot usage, payment success rates, and integration health to support data-informed decision-making.

**Junior Portfolio Analyst | Python Developer | Contract**, ADEI (PTY) Ltd Mar 2025 – Nov 2025

- Built end-to-end Python automation frameworks, including SACRRA-compliant fixed-width file submissions, API integrations, WhatsApp bots, and standalone executables with scheduling, logging, and error handling
- Designed and deployed multiple Power BI dashboards delivering real-time operational, compliance, and

financial reporting with automated refresh and actionable insights

- Developed robust SQL Server data pipelines using Python (pyodbc), stored procedures, and ETL processes to extract, transform, and load data from APIs, Excel, and CSV sources
- Created intelligent data validation and automation solutions (e.g., regex checks, scientific notation detection, resubmission controls) to ensure accuracy, compliance, and efficiency in reporting workflows

**Data Scientist | Internship**, Sand Technologies

Sep 2024 – Nov 2024

- Developed a Python-based lead conversion model for an insurance client, achieving 0.800 accuracy (improved to 0.802 with hyperparameter tuning)
- Visualized insights with Power BI dashboards
- Collaborated via GitHub
- Managed scalable data storage on AWS

**Master Data Junior Officer | Contract**, Genpact

Nov 2023 – Jan 2024

- Utilized SQL to verify outlet folder completeness
- Ensured proper documentation and contract signing
- Tracked progress in Excel for 600 outlet folders daily
- Collaborated on process improvements to streamline organization and ensure compliance

**Master Data Analyst | Contract**, ERG Africa

Dec 2022 – Aug 2023

- Extracted part numbers and MFR/OEM data for 350 products daily using Excel and SQL
- Performed QA to ensure accuracy
- Developed Power BI dashboards for performance tracking
- Collaborated on process optimizations to enhance workflows and data quality.

## Projects

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**Predictive Model for Loan Approval Status**

[GitHub Repo](#)

- Developed a predictive model to forecast loan application approval status using demographic and financial features.
- Streamlined the approval process, making it more efficient and data-driven
- Reduced default risks by identifying high-risk applications early.
- Provided valuable insights for both lenders and applicants to make informed decisions

**NLP-Based News Article Classification**

[GitHub Repo](#)

- Developed and deployed an automated news article classification system using machine learning techniques.
- Accurately categorized articles into predefined groups, improving content organization.
- Enhanced operational efficiency and improved reader satisfaction for a news outlet.

**Anime Rating Prediction and Recommendation System**

[GitHub Repo](#)

- Built an application that provides predicted anime ratings and personalized recommendations based on clients' historical preferences.

## Skills Summary

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**Languages:** Python, SQL, C++ , Java

**Automation & Integration:** Flask, Redis, PyODBC, Requests, OpenPyXL, PyInstaller

**BI & Analytics Tools:** Power BI, Excel, SAS, SPSS

**Data & ML Libraries:** Pandas, NumPy, Scikit-Learn, MLflow, Matplotlib, Seaborn

**Databases & Cloud:** MySQL, SQLite, SQL Server, AWS

**Dev & Collaboration:** Visual Studio Code, Jupyter Notebook, GitHub, Trello, Confluence, Azure DevOps

## Certificates

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Simplilearn, Business Intelligence Fundamentals

[Certificate](#)

## References

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References available on request